The `mystyletau` package

Antonello Meccariello & Francesco Meccariello

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1 Introduction

The `mystyletau` package introduces two custom glyphs for the letter τ. While the standard notation for a topological space (X, τ_X) is functional, the authors believe these variants offer superior aesthetic appeal for denoting topologies or algebraic structures.

2 New in Version 2.0

Version 2.0 marks a significant technical shift. The symbols are no longer external PDF images; they are now implemented as true vector glyphs within a dedicated TrueType font (`mytautwo.ttf`). This ensures:

- **Mathematical Integration:** Symbols scale perfectly in subscripts ($\mathscr{\tau}_{\mathscr{\tau}}$) and superscripts.
- **Color Support:** Symbols dynamically respond to color commands, e.g., $\mathscr{\tau}$ or $\mathscr{\tau}$.
- **Mode Independence:** Commands work in both text and math mode via `\ensuremath`.

3 Usage

> **Important:** This package requires the use of **LuaLaTeX** or **XeLaTeX**. Compilation with `pdflatex` will result in a fatal error, as the `fontspec` package is used to handle the custom vector font.

The following commands provide access to the custom glyphs:

- `\vtau`: $\mathscr{\tau}$
- `\atau`: $\mathscr{\tau}$

4 Examples

Consider the following statements to compare the visual impact:

1. Standard: Let (X, τ_X) be a topological space.
2. Variant A: Let $(X, \mathscr{\tau}_X)$ be a topological space.
3. Variant B: Let $(X, \mathscr{\tau}_X)$ be a topological space.

The authors suggest that notations like $(X, \mathscr{\tau}_X)$ and $(X, \mathscr{\tau}_X)$ provide a more elegant visual impact compared to the standard Computer Modern (X, τ_X).